NAP+

Material Safety Data Sheet

NFPA Classification	DOT / TDG Pictograms	WHMIS Classification	PROTECTIVE CLOTHING
Health 1, Flammability 0, Reactivity 0, Specific Hazard			

Section I. Chemical Product and Company Identification

PRODUCT NAME/ TRADE NAME	Power Plus ST

SYNONYM		MSDS NUMBER:	PPST 1
CHEMICAL NAME	Metals EthyleneDiamine TetraAcetate plant micronutrients.	REVISION NUMBER	01.0
CHEMICAL FAMILY	Synthetic Sequesterant	MSDS prepared by the Environment, Health and Safety Department on:	January 23 , 2024
CHEMICAL FORMULA	C10H12MetalN2O8.Metal C4H9NO3	24 HR EMERGENCY TELEPHONE NUMBER:	
MATERIAL USES	Agricultural use: Fertilizer ingredient.	Trans portation E mergency: + (662) 752-6959 Medical E mergency: + (662) 320-4100	

MANUFACTURER

napnutriscience (Thailand) Company Limited
613/1 Bangpoo Industrial Estate, Moo 4
Praksa, Amphoe Muang, Samuthprakarn 10280,
THAILAND

DISTRIBUTOR

napnutriscience Company Limited
626 Bangpoo Industrial Estate, Moo 4
Praksa, Amphoe Muang, Samuthprakarn 10280,
THAILAND

Section II. Hazardous Ingredients

		Exposure Limits (ACGIH)				
NAME	CAS #	CAS #	STEL mg/m^3	STEL ppm	% by Weight	% by Weight
Zinc Sulphate Monohydrate Magnesium Sulphate Monohydrate L-Threonine	7446-19-7 14168-73-1 72-19-5					7.0 as Zn , 5.0 as Mg , 7.0 as L-Amino

TOXICOLOGICAL DATA ON INGREDIENTS

- Magnesium Sulphate Monohydrate
ORAL:(LD50): Acute: 2000mg/kg(RAT)
- Zinc Sulphate Monohydrate
- ORAL:(LD50):Acute 2949mg/kg(RAT)
- L-Threonine
- ORAL:(LD50)Acute: 3098 mg/kg(RAT)

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Section III. Hazards Identification.

POTENTIAL ACUTE HEALTH EFFECTS	Will cause irritation of the digestive tract if swallowed. May cause severe eye irritation. May cause skin irritation. Over-exposure by inhalation may cause respiratory tract irritation.
POTENTIAL CHRONIC HEALTH EFFECTS	CARCINOGENIC EFFECTS: NONE by ACGIH, EPA, IARC, NTP, OSHA. MUTAGENIC EFFECTS: NONE by ACGIH, EPA, IARC, NTP, OSHA. TERATOGENIC EFFECTS: NONE by ACGIH, EPA, IARC, NTP, OSHA.

Section IV. First Aid Measures

EYE CONTACT	Immediately flush eyes with water for at least 15 minutes. Use warm water if available. Get medical attention if irritation persists.
MINOR SKIN CONTACT	After skin contact wash immediately with plenty of water and non-abrasive soap. Cover irritated skin with a good quality skin cream. If irritation persists, seek medical attention. Wash contaminated clothing before reusing.
EXTENSIVE SKIN CONTACT	No additional information.
MINOR INHALATION	Allow to rest in a well ventilated area. Seek medical attention, if not feeling well.
SEVERE INHALATION	No additional information.
SLIGHT INGESTION	Irritating to the mouth, throat, stomach, digestive tract and bowels resulting in nausea, vomiting and diahrrea. Have affected person drink several glasses of water to assist in purging system. If the person is conscious, induce vomiting. Seek medical attention.
EXTENSIVE INGESTION	No additional information.

Section V. Fire and Explosion Data

THE PRODUCT IS	Non-flammable.
AUTO-IGNITION TEMPERATURE	Not applicable.
FLASH POINT	Not applicable.
FLAMMABILITY LIMITS	Not applicable.
PRODUCTS OF COMBUSTION	Not applicable.
FIRE HAZARD IN THE PRESENCE OF VARIOUS SUBSTANCES	Not applicable.
EXPLOSION HAZARD IN THE PRESENCE OF VARIOUS SUBSTANCES	This substance is non-explosive.
FIRE FIGHTING MEDIA AND INSTRUCTIONS	Material will not burn. Undergoes thermal decomposition at elevated temperatures to release toxic gases. Fires involving this material may be extinguished by any means consistent with surrounding materials. As in any fire, wear MSHA/NIOSH approved pressure-demand self-contained breathing apparatus, or equivalent and full protective gear.
SPECIAL REMARKS ON FIRE HAZARDS	A self contained breathing apparatus should be used to avoid inhalation of toxic fumes.
SPECIAL REMARKS ON EXPLOSION HAZARDS	No additional information.

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Section VI. Accidental Release Measures

SPILL	Use appropriate tools or equipment to place the spilled solid in a suitable container for reuse or disposal. Flush or wash the contaminated surface. Recover wash water and dispose of according to local and regional authority requirements.
	Stop leak if without risk.

Section VII. Handling and Storage

PRECAUTIONS	Avoid contact with skin and eyes. DO NOT breathe dust. After handling, always wash hands thoroughly with soap and water. Use process enclosures, local exhaust ventilation, or other engineering controls to keep airborne levels below recommended exposure limits.
STORAGE	Keep container tightly closed. Store in a dry, cool and well ventilated area. Keep away from food, drink and animal feeds. Keep out of reach of children.

Section VIII. Exposure Controls/Personal Protection

ENGINEERING CONTROLS	Use process enclosures, local exhaust ventilation, or other engineering controls to keep airborne levels below recommended exposure limits. If user operations generate dust, fumes or mist, use adequate local exhaust or general ventilation to keep exposure to airborne contaminants below the exposure limits.
PERSONAL PROTECTION	The selection of personal protective equipment varies, depending upon conditions of use. Under well controlled conditions where no direct contact with the substance exists and exposures are below the occupational exposure limit, normal work clothing may suffice. However, protective clothing and appropriate respiratory protection must be available in the area in the event of emergency. Wear appropriate respirator when ventilation is inadequate. Where skin and eye contact may occur as a result of brief periodic exposures, wear clothing with long sleeves, coverals, chemical resistant gloves, impervious chemical resistant jacket and pants, and safety glasses with side shields. A respiratory protection program that meets OSHA 29 CFR 1910.134 requirements must be followed whenever workplace conditions warrant a respirator's use.
PERSONAL PROTECTION IN CASE OF LARGE RELEASE	No further information.
EXPOSURE LIMITS	ACGIH TLV-TWA 5mg/m^3 Fed OSHA Permissible Exposure Limit: Table Z-1 8-hr Time Weighted Avg: 15 mg/m^3 as total dust MI OSHA Permissible Exposure Limit: 10 mg/m^3

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Federal, State, and Provincial exposure limits may vary. Consult local officials for acceptable exposure limits in your jurisdiction.

Section IX. Physical and Chemical Properties

PHYSICAL STATE AND APPEARANCE	Crystalline fine powder		
MOLECULAR WEIGHT	Not available.	COLOR	White
pH (1% SOLN/WATER)	2.0 - 4.0	ODOR	Sulfur (Slight)
BOILING POINT	Not available.	ODOR THRESHOLD	Not available.
MELTING POINT	256 C (decompose)	TASTE	Bitter
CRITICAL TEMPERATURE	Not available.	VOLATILITY	Not available.
SPECIFIC GRAVITY g/cc	1.10-1.20	SOLUBILITY	Soluble in cold water, hot water, methanol.
BULK DENSITY kg/m^3	1100-1200 kg/m^3	DISPERSION PROPERTIES	Not available.
VAPOR PRESSURE	Not available.	WATER/OIL DIST. COEFF.	Not available.
VAPOR DENSITY	Not available.		

Section X. Stability and Reactivity Data

STABILITY	The product is stable.
INSTABILITY TEMPERATURE	Not available.
CONDITIONS OF INSTABILITY	No additional information.
INCOMPATABILITY WITH VARIOUS SUBSTANCES	Highly reactive with oxidizing agents. Slightly reactive with moisture.
CORROSIVITY	Highly corrosive in the presence of steel, or copper. Corrosive in presence of aluminum or 304 stainless steel. Slightly corrosive in the presence of 316 stainless steel.
SPECIAL REMARKS ON REACTIVITY	Avoid strong oxidizing agents.
SPECIAL REMARKS ON CORROSIVITY	Contact your sales representative or metallurgical specialist to ensure compatibility with system equipment.

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Section XI. Toxicological Information

SIGNIFICANT ROUTES OF EXPOSURE	Eye contact. Inhalation. Ingestion.
TOXICITY TO ANIMALS	See Section II.
SPECIAL REMARKS ON TOXICITY TO ANIMALS	May be harmful to livestock and wildlife if ingested. Clean up all spilled material, especially where bulk fertilizer loading of equipment occurs to prevent animal exposure. The product itself and its products of degradation are not harmful under normal conditions of use.
OTHER EFFECTS ON HUMANS	No additional information.
SPECIAL REMARKS ON CHRONIC EFFECTS ON HUMANS	No additional remark.
SPECIAL REMARKS ON OTHER EFFECTS ON HUMANS	No additional remark.

Section XII. Ecological Information

ECOTOXICITY	Low toxicity for humans or animals under normal conditions of use. May be harmful to livestock and wildlife if ingested. Clean up all spilled material.
BOD and COD	Not available.
PRODUCTS OF DEGRADATION	Possibly hazardous short term degradation products are not likely. However, long term degradation products may arise.
TOXICITY OF THE PRODUCTS OF DEGRADATION	The products of degradation are less toxic than the product itself.
SPECIAL REMARKS ON THE PRODUCTS OF DEGRADATION	No additional remark. Reclaiming material may not be viable.

Section XIII. Disposal Considerations

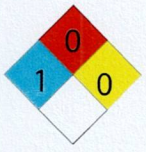
WASTE DISPOSAL OR RECYCLING	Recover spilled material and use for intended purpose.

Section XIV. Transport Information

DOT / TDG CLASSIFICATION	Not controlled under TDG (Canada) or D.O.T. (U.S.A.) or IATA
PIN and Shipping Name	Not applicable.
SPECIAL PROVISIONS FOR TRANSPORT	No additional remark.
DOT (U.S.A) (Pictograms)	

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Section XV. Other Regulatory Information and Pictograms

OTHER REGULATIONS	OSHA: Hazardous by definition of Hazard Communication Standard (29 CFR 1910.1200). TSCA (Toxic Substance Control Act): This product is listed on the TSCA Inventory.	
OTHER CLASSIFICATIONS	HCS (U.S.A.)	HCS: Irritating substance. May be toxic for the nervous system.
	DSCL (EEC)	R25- Toxic if swallowed. R36/37/38- Irritating to eyes, respiratory system and skin.
National Fire Protection Association (U.S.A.)	Hazards presented under acute emergency conditions only:	 Health Fire Hazard Reactivity Specific Hazard
TDG (Pictograms - Canada)		
DSCL (Europe) (Pictograms)	Not Available No Disponible Pas Disponible	
ADR (Europe) (Pictograms)	Not Available No Disponible Pas Disponible	

Section XVI. Other Information

REFERENCES

-Transportation of Dangerous Goods Act and Clear Language Regulations, current revision.
-Canada Gazette Part II, Vol. 122, No. 2 Registration SOR/88-64 31 December, 1987 Hazardous Products Act "Ingredient Disclosure List".
-Domestic Substances List, Canadian Environmental Protection Act.
-29 CFR Part 1910
-33 CFR Parts 151, 153, 154, 156
-40 CFR Parts 1-799
-46 CFR Part 153
-49 CFR Parts 1-199
-American Conference of Governmental Industrial Hygienists, Threshold Limit Values for Chemical Substances, 2003.
-NFPA 704, National Fire Codes Online, National Fire Protection Association, current edition at time of MSDS preparation.
-Corrosion Data Survey, Sixth Edition, 1985, National Association of Corrosion Engineers
-TOMES® System: Heitland G & Hurlbut KM (Eds) (electronic version): MICROMEDEX, Greenwood Village, Colorado, USA. Available at: http://csi.micromedex.com (2004). The TOMES® System includes MEDITEXT® Medical Management; HAZARDTEXT® Hazard Management; INFOTEXT® Documents; ERG2000 Emergency Response Guidebook Documents; REPROTEXT®: Heitland G & Hurlbut KM (Eds); CHRIS Hazardous Chemical Data:

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	U.S. Department of Transportation, U.S. Coast Guard, Washington, D.C. (2004); HSDB: Hazardous Substances Data Bank. National Library of Medicine, Bethesda, Maryland (2004); IRIS: Integrated Risk Information System. U.S. Environmental Protection Agency, Washington, D.C. (2004); NIOSH: Pocket Guide to Chemical Hazards. National Institute for Occupational Safety and Health, Cincinnati, Ohio (2004); OHM/TADS: Oil and Hazardous Materials Technical Assistance Data System. U.S. Environmental Protection Agency, Washington, D.C. (2004); REPROTOX®: Scialli A.R. Georgetown University Medical Center and Reproductive Toxicology Center, Columbia Hospital for Women Medical Center, Washington, D.C. (2004); RTECS®: Registry of Toxic Effects of Chemical Substances. National Institute for Occupational Safety and Health, Cincinnati, Ohio (2004); and SHEPARDS: Shepard T.H.: Shepard's Catalog of Teratogenic Agents (2004). -Michigan Office of Regulatory Reform R325.51101et seq
OTHER SPECIAL CONSIDERATIONS	No additional information.
FOR FURTHER SAFETY, HEALTH, OR ENVIRONMENTAL INFORMATION ON THIS PRODUCT, CONTACT	napnutriscience Company Limited Environment, Health and Safety Department Fax (+662) 320-4100

NOTICE TO READER

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